Question	Answer	Marks	AO	Guidance
13	$2y\frac{dy}{dx}$	B1	1.1	chain rule
	$y + x\frac{dy}{dx}$	B1	3.1a	product rule
	(their previous terms) $+ 2x - 1 = 0$	B1	1.1	may award if "= 0" seen later, but not if RHS is $\frac{dy}{dx}$
	their $y + 2x - 1 = 0$	M1	2.1	substitution of $\frac{dy}{dx} = 0$; may follow (incorrect) rearrangement; dependent on award of at least one **B** mark **NB** $\frac{dy}{dx} = \frac{1-2x-y}{2y+x}$
	$(1-2x)^2 + x(1-2x) + x^2 - x = 1$ **or** $y^2 + \frac{(1-y)y}{2} + \frac{(1-y)^2}{4} - \frac{1-y}{2} = 1$	M1	3.1a	elimination of x or y using expression or value obtained from use of $\frac{dy}{dx} = 0$; dependent on award of at least one **B** mark
	$3x^2 - 4x[= 0]$ **or** $3y^2 + 2y - 5[= 0]$	A1	1.1	
	$x = 0, x = \frac{4}{3}$	M1	1.1	values of x **or** y found from their quadratic
	$y = 1, y = -\frac{5}{3}$	M1	1.1	values of y **or** x found from substitution of both x **or** both y values; must see substitution unless values correct **NB** may see extra points $y = -1$ or $\frac{1}{3}$ from substitution into original equation
	(0,1) and $\left(\frac{4}{3}, -\frac{5}{3}\right)$ **or** $x = 0, y = 1$ and $x = \frac{4}{3}, y = -\frac{5}{3}$	A1	3.2a	**A0** if extra points in final answer; dependent on fully correct working throughout; if **M0M0** allow **SCB1** for 1 correct pair of coordinates and no others
		[9]		

Question	Answer	Marks	AO	Guidance
	Alternatively $2x\frac{dx}{dy} - \frac{dx}{dy}$	**B1**		chain rule
	$x + y\frac{dx}{dy}$	**B1**		product rule
	(their previous terms) $+2y = 0$	**B1**		may award if "= 0" seen later, but not if RHS is $\frac{dy}{dx}$
	their $1 - 2x - y = 0$	**M1**		from setting denominator of $\frac{dx}{dy} = \frac{2y+x}{1-2x-y}$ equal to 0 or rearranging to find $\frac{dy}{dx} = \frac{1-2x-y}{2y+x}$ and setting equal to 0; dependent on award of at least one **B** mark
	$(1-2x)^2 + x(1-2x) + x^2 - x = 1$ or $y^2 + \frac{(1-y)y}{2} + \frac{(1-y)^2}{4} - \frac{1-y}{2} = 1$	**M1**		elimination of x or y using expression or value obtained from use of $\frac{dy}{dx} = 0$; dependent on award of at least one **B** mark
	$3x^2 - 4x[= 0]$ **or** $3y^2 + 2y - 5[= 0]$	**A1**		
	$x = 0, x = \frac{4}{3}$	**M1**		values of x **or** y found from their quadratic
	$y = 1, y = -\frac{5}{3}$	**M1**		values of y **or** x found from substitution of both x **or** both y values; must see substitution unless values correct **NB** may see extra points $y = -1$ or $\frac{1}{3}$ from substitution into original equation
	(0,1) and $\left(\frac{4}{3}, -\frac{5}{3}\right)$ **or** $x = 0, y = 1$ and $x = \frac{4}{3}, y = -\frac{5}{3}$	**A1**		**A0** if extra points in final answer; dependent on fully correct working throughout if **M0M0** allow **SCB1** for 1 correct pair of coordinates and no others
		9		